

Fred K. Gruber

Senior Principal Scientist

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in [fredgruber](#)

Summary

With 10 years of experience in analyzing medical datasets from clinical and real-world sources, I have a proven track record in the investigation of prognostic and predictive biomarkers as well as drug target discovery. I am proficient in all stages of data analysis, from preprocessing to modeling and interpretation, and have successfully led multiple projects with top pharmaceutical scientists as stakeholders. My expertise lies in statistical and machine learning models applied to medical data, with a focus on causal methods, high-dimensional variable selection, data integration, and interpretability. I am skilled in working with multi-modal high-dimensional genomic data and clinical variables, utilizing techniques such as bayesian networks, logistic regression, survival analysis, neural networks, and more. Additionally, I am the developer and maintainer of multiple internal R packages for data preprocessing, visualization, machine learning, and analysis as well as maintainer of the `simcausal` R package for simulation of causal networks.

Professional Experience

Aug 2018 – **Senior Principal Scientist, Aitia, Somerville, MA**

current

- Analysis of high-dimensional multi-modal biological data.
- Experience with RNAseq, simple nucleotides variants, structural variants, copy number variants, and clinically relevant variables.
- Focus on discovering biomarkers and drug targets for different diseases.
- Experience in analyzing oncological datasets for lung cancer, multiple myeloma, prostate cancer, pancreatic cancer, and kidney cancer.
- Utilization of different data sources such as TCGA/GDC, Flatiron, ICGC, Tempus MMRF, and others.
- Involvement in all stages of the analysis process, from data preprocessing to model construction and interpretation as well as presentation to clients.
- Use of proprietary causal Bayesian network learning and predictive tools as main tools.
- Exploration of open source machine learning techniques applicable to datasets.
- Investigation of unsupervised machine learning methods for tumor subclonal composition analysis and patient drug response stratification.
- Developed the best performing model predicting high-risk multiple myeloma in the Multiple Myeloma DREAM Challenge finding a novel expression-based risk marker, PHF19, which was further validated in vitro and published in journal Leukemia.

Oct 2016 – Principal Scientist, Aitia, Cambridge, MA

- Oct 2018**
- Responsible for the development of causal network models for multiple myeloma based on multiple iterations of the CoMMPass MMRF dataset.
 - Developed preprocessing pipelines to prepare diverse data modalities including copy number variants, structural variants, single nucleotide variants, gene expression, and clinical data.
 - Validation of result by literature search.
 - Published multiple peer-reviewed conference papers in a major blood cancer conference.

Jul 2013 – Senior Systems Biologist, Aitia, Cambridge, MA

- Oct 2016**
- Developed, implemented, and benchmarked algorithms for the prediction of rare adverse events on patients taking the client's drug.
 - Developed novel algorithms implementing variations of logistic regression that perform better under imbalanced outcomes.
 - Comparative analysis using 17 public-domain datasets
 - Applied to the client's clinical and genomic multiple sclerosis dataset.
 - Successfully delivered R package with new algorithm.
 - Success of algorithm lead to production code in the internal causal and predictive tools at Aitia.
 - Developed predictive and causal models of Chron's disease for the purpose of elucidating the relationships between TL1A, DR3, IFN γ , and a variety of genetic, gene expression, immune response, and clinical variables from tissue samples.

May 2009 – Postdoctoral Scientist, Schlumberger-Doll Research Center, Cambridge, MA

- Jun 2013**
- Designed, implemented, and tested algorithms for analyzing Nuclear Magnetic Resonance measurements for oil exploration
 - Utilized Matlab extensively for research purposes
 - Utilized Python for production code integration with company software
 - Worked with optimization algorithms, sparse coding, regularization, and general machine learning tools.

Education

2004 – 2009 Ph.D., Electrical engineering, Northeastern University, Boston, MA

- Research Assistant at Northeastern University's Communications and Digital Signal Processing Center.
- Member of the Honor Society of Eta Kappa Nu.
- GPA of 3.9/4.0
- Developed algorithms for the determination of the location of objects from the measure of scattered fields. This is a fundamental problem in the area of remote sensing and inverse problems. Also studied the fundamental limits of communication using electromagnetic fields.

2003 – 2004 Master of science, Industrial Engineering and Management Systems, University of Central Florida, Orlando, FL

- Research Assistant the Center for NASA Simulation Research.
- Phi Kappa Phi honor society
- GPA of 3.9/4.0
- Recipient of the University of Central Florida's Graduate Merit Fellowship Award, 2004
- Worked at the NASA Simulation Research Group and was involved in the design of an advanced simulation of the probability of failure a space shuttle during reentry and launch.
- For my master thesis, I developed a Genetic Algorithm in C++ for the optimal selection of the parameters of a Support Vector Machine. Published this work in book chapter.

1998 – 2003 Bachelors, Electrical and Electronic Engineering, Universidad Tecnológica de Panamá, Panamá City, Panamá

- Member of IEEE
- GPA of 3.74/4.0
- Honor Scholarship for undergraduate studies in Engineering at the Technological University of Panama, 1998 - 2002.
- Gold Medal "Guillermo Andreve" for educational achievements, Panama, 1997.

Relevant Certificates

- 2023 Combining information for Causal Inference, Harvard T.H. Chan School of Public Health, Cambridge, MA**
- 2022 Target Trial Emulation, Harvard T.H. Chan School of Public Health, Cambridge, MA**
- 2020 Applied Deep Learning Boot Camp, MIT Professional School, Cambridge, MA**
- 2018 An Introduction to Causal Inference, Harvard T.H. Chan School of Public Health, Cambridge, MA**
- 2016 Data Science: Data to insights, MIT Professional School, Cambridge, MA**

Skills

- R
- ggplot2
- shiny
- python
- matlab
- elisp
- machine learning
- causal discovery
- causal inference
- python
- High-dimensional data analysis
- AWS
- data wrangling
- data analysis
- data visualization
- statistics
- teamwork
- scientific presentation
- research

Languages

- English Fluent
- Spanish Native

Selected Publications

- [1] Brandon Nathasingh, **Fred K. Gruber**, Derek Walkama, Jeanne Latourelle, Bruce W Church, et al. “Multiple Myeloma Causal Model Suggests 1q21 Amplification and 13q14 Deletion Co-Alteration Interact to Influence Patient Prognosis”. In: *65TH ASH ANNUAL MEETING AND EXPOSITION*. Dec. 2023.
- [2] Boris Hayete, **Fred K. Gruber**, Anna Decker, and Raymond Yan. “Identification of Latent Variables From Graphical Model Residuals”. In: *arXiv:2101.02332 [cs, q-bio, stat]* (Jan. 2021). arXiv: 2101.02332 [cs, q-bio, stat]. URL: <http://arxiv.org/abs/2101.02332> (visited on 02/08/2021).
- [3] Likhitha Kolla, **Fred K. Gruber**, Omar Khalid, Colin Hill, and Ravi B. Parikh. “The Case for AI-driven Cancer Clinical Trials – The Efficacy Arm in Silico”. In: *Biochimica et Biophysica Acta (BBA) - Reviews on Cancer* 1876.1 (Aug. 2021), p. 188572. ISSN: 0304419X. DOI: 10.1016/j.bbcan.2021.188572. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0304419X2100069X> (visited on 06/28/2021).
- [4] Mike J. Mason, Carolina Schinke, Christine L. P. Eng, Fadi Towfic, **Fred K. Gruber**, et al. “Multiple Myeloma DREAM Challenge Reveals Epigenetic Regulator PHF19 as Marker of Aggressive Disease”. In: *Leukemia* (Feb. 2020), pp. 1–9. ISSN: 1476-5551. DOI: 10.1038/s41375-020-0742-z. URL: <https://www.nature.com/articles/s41375-020-0742-z> (visited on 02/17/2020).
- [5] Michael J. Mason, Carolina D. Schinke, Christine Eng, Fadi Towfic, **Fred K. Gruber**, et al. “Crowdsourced High-Risk Classifiers for Multiple Myeloma Patients Commonly Identify PHF19 As a Robust Progression Biomarker”. In: *Blood* 134.Supplement_1 (Nov. 2019), pp. 4370–4370. ISSN: 0006-4971. DOI: 10.1182/blood-2019-127362. URL: https://ashpublications.org/blood/article/134/Supplement_1/4370/424531/Crowdsourced-High-Risk-Classifiers-for-Multiple (visited on 08/25/2020).
- [6] Arnold Bolomsky, **Fred K. Gruber**, Kathrin Stangelberger, Leon Furchtgott, Dominik Arnold, et al. “Preclinical Validation Studies Support Causal Machine Learning Based Identification of Novel Drug Targets for High-Risk Multiple Myeloma”. In: *Blood* 132.Supplement 1 (Nov. 2018), pp. 3210–3210. ISSN: 0006-4971. DOI: 10.1182/blood-2018-99-117886. URL: <https://ashpublications.org/blood/article/132/Supplement%201/3210/263718/Preclinical-Validation-Studies-Support-Causal> (visited on 08/25/2020).
- [7] **Fred K. Gruber**. “Evolutionary Optimization of Support Vector Machines Using Genetic Algorithms”. In: *Artificial Intelligence: Advances in Research and Applications*. Ed. by Luis Rabelo, Sayli Bhide, and Edgar Gutierrez. COMPUTER SCIENCE, TECHNOLOGY AND APPLICATIONS. Nova, 2018.
- [8] Leon Furchtgott, Arnold Bolomsky, **Fred K. Gruber**, Mehmet Kemal Samur, Jonathan J. Keats, et al. “Multiple Myeloma Drivers of High Risk and Response to Stem Cell Transplantation Identified By Causal Machine Learning: Out-of-Cohort and Experimental Validation”. In: *Blood* 130.Supplement 1 (Dec. 2017), pp. 3029–3029. ISSN: 0006-4971. DOI: 10.1182/blood.V130.Suppl_1.3029.3029. URL: <https://ashpublications.org/>

blood/article/130/Supplement%201/3029/80484/Multiple-Myeloma-Drivers-of-High-Risk-and-Response (visited on 08/25/2020).

- [9] **Fred K. Gruber**, Jonathan J. Keats, Kyle McBride, Karl Runge, Diane Wuest, et al. “Bayesian Network Models of Multiple Myeloma: Drivers of High Risk and Durable Response”. In: *Blood* 128.22 (Dec. 2016), pp. 4406–4406. ISSN: 0006-4971. DOI: 10.1182/blood.V128.22.4406.4406. URL: <https://ashpublications.org/blood/article/128/22/4406/101366/Bayesian-Network-Models-of-Multiple-Myeloma> (visited on 08/25/2020).
- [10] **Fred K. Gruber**, Boris Hayete, Jonathan J Keats, Kyle McBride, Runge Karl, et al. “Investigation of Mechanisms of Response in Multiple Myeloma Via Bayesian Causal Inference: An Early Analysis of the Commpass Study Data”. In: *BLOOD* 126.23 (2015).